

Joshua Faskowitz

Postdoctoral Fellow

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📍 Bethesda, MD

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Education

PhD	Indiana University Bloomington , Neuroscience and Psychology	Bloomington, IN
	• Advisor: Dr. Olaf Sporns • Committee: Dr. Olaf Sporns, Dr. Aina Puce, Dr. Amanda Mejia, and Dr. Richard Betzel	2016 – 2021
BA	University of Southern California , Neuroscience and Cognitive Science	Los Angeles, CA
		2010 – 2014

Experience

National Institute of Mental Health , Postdoctoral Fellow, Section on Functional Imaging Methods	Bethesda, MD
PI: Dr. Peter Bandettini	2022 – present
• Develop and apply methods for characterizing fMRI correlation structure across space and time. • Study how preprocessing, physiology, and signal interpretation shape inferences about functional brain organization.	4 years
IU Department of Psychological and Brain Sciences , Postdoctoral Researcher	Bloomington, IN
PIs: Dr. Olaf Sporns and Dr. Richard Betzel	2021 – 2022
• Extended edge-centric and community-based network models for studies of human brain organization. • Supported collaborative neuroimaging and network science projects through analysis, mentoring, and shared computational workflows.	1 year
IU Computational Cognitive Neuroscience Lab , Graduate Student	Bloomington, IN
PI: Dr. Olaf Sporns	2016 – 2021
• Investigated mesoscale organization, edge-centric representations, and time-varying structure in structural and functional brain networks. • Collaborated across labs on methods, data sharing, and reproducible computational analyses in network neuroscience.	5 years
USC Imaging Genetics Center , Project Assistant	Los Angeles, CA
PI: Dr. Paul Thompson; under supervision of Dr. Neda Jahanshad	2014 – 2016
• Processed MRI data into analysis-ready neuroimages by designing reproducible workflows.	2 years
USC Emotion and Cognition Lab , Undergraduate Research Assistant	Los Angeles, CA
PI: Dr. Mara Mather; under supervision of Dr. Allison Ponzio	2012 – 2014
• Administered experiments over five semesters and one summer, totaling more than 800 lab hours.	2 years

Research Areas

Focus: Network neuroscience; network science algorithms and applications; community detection; edge-centric modeling; fMRI analysis and interpretation; time-varying connectivity

Honors and Awards

Irving J. Saltzman Award

Departmental award for an outstanding recent PhD graduate.

IU Department of Psychological and Brain Sciences
2023

Linda and Jack Gill Graduate Student Award, Honorable Mention

One of four awardees selected from regional neuroscience programs at Indiana University, IUPUI, Purdue, and Notre Dame.

IU Gill Center
2021

J. R. Kantor Graduate Award

Departmental award for an outstanding advanced graduate student, shared with another student.

IU Department of Psychological and Brain Sciences
2021

Dissertation Research Fellowship

Fellowship support totaling \$20,000.

IU College of Arts and Sciences
2020-2021

ACNN Travel Award

Travel award of approximately \$300.

Advanced Computational Neuroscience Network Workshop
2016, 2019

PNS Travel Award

Travel award of \$800.

Network Neuroscience satellite at NetSci
2019

Graduate Research Fellowship Program

NSF GRFP support at \$34,000 per year for three years.

National Science Foundation
2016-2021

Rebec Fellow

Fellowship award of \$4,000.

IU Department of Psychological and Brain Sciences
2016

Trustee Scholarship

Full-tuition scholarship awarded for academics, leadership, and community service.

University of Southern California
2010-2014

Order of Troy

Award for academic and leadership excellence; one of 112 undergraduate awardees.

University of Southern California
2014

Renaissance Scholar and Discovery Scholar

Recognition for excelling in widely separated fields of study and for conducting scholarly research.

University of Southern California
2014

Invited Talks

Network Outcomes, Uses, and Null Models: Spring 2022; Practical Applications of Network Neuroscience Workshop; co-organized with Dr. Jenya Chumin

Brain Networks: Close to the Edge: Spring 2022; Washington University in St. Louis, Neuroimaging in Health and Disease Seminar Series

Brain Networks: Close to the Edge: Spring 2022; Dr. Lucina Uddin Lab Meeting

fMRI Preprocessing Workshop: Summer 2022; Indiana University; three-day workshop covering workflows for processing functional neuroimaging data on IU high-performance computing resources

Publicly Available Neuroimaging Databases: Spring 2020; CUNY 2020 Workshop on Remote Data Collection

Teaching

The Consciousness Network: Summer 2023 and Summer 2024; NIH summer journal club; co-organized with Dr. Sharif Kronemer

P303: Health Psychology: Spring 2019; Indiana University; teaching assistant

P211: Methods of Experimental Psychology: Fall 2018; Indiana University; instructor of record

Workshops and Hackathons

NIMH Grant Writing Workshop: 2024 and 2025; National Institute of Mental Health; full-day overview of grantsmanship

Brainhack-Networks: May 2019; University of Vermont; pre-NetSci 2019 workshop

Brainhack Global at IU: Spring 2018; Indiana University

5th Indiana Neuroimaging Symposium and Hackathon: Fall 2017; Purdue University

Neurohackweek: Fall 2017; University of Washington eScience Institute

4th Indiana Neuroimaging Symposium and Hackathon: Fall 2016; Indiana University

OHBM Hackathon: June 2015; Honolulu, USA; pre-OHBM 2015 workshop

Open Science

Multi Atlas Transfer Tools: Fit parcellations using FreeSurfer and distributed as a Brainlife.io application; run more than 19,000 times as a Brainlife.io app

fMRI-2-Mat Tools: Tools for nuisance regression and matrix-based downstream analysis of fMRI data

Lifespan structural connectivity matrices from Faskowitz (2018): Shared dataset released for downstream reuse

Brain networks across the web: Curated collection of openly available brain-network datasets

Skills

Coding and scripting

Computational tools

Neuroimaging packages

Professional Service

Journal peer review: Aperture Neuro; Brain Imaging and Behavior; Brain Structure and Function; Biological Psychiatry; Cerebral Cortex; Communications Biology; Human Brain Mapping; Nature Communications; Network Science; Network Neuroscience; NeuroImage; Scientific Data; Scientific Reports; The Journal of Neuroscience; Trends in Cognitive Sciences

Professional Affiliations

- Network Science Society
- Organization for Human Brain Mapping
- Society for Neuroscience

Scholarly Contributions

Publications: For a list of articles, visit my Google Scholar or the publications page.